

Predicting Vehicle Loan Defaults Using Explainable Machine Learning Models

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Abstract—Predicting loan default is an important task for financial institutions, yet it becomes challenging when the number of defaulters is much smaller than the number of non-defaulters. This study compares several machine learning approaches for vehicle loan default prediction using the L&T dataset, which contains 233,154 loan records described by 41 features. The default rate is only 21.7%, making class imbalance a major challenge. The analysis included data cleaning, feature engineering, exploratory data analysis, feature selection, predictive modeling, and borrower segmentation through clustering. Logistic regression, decision tree, support vector machine, random forest, XGBoost, LightGBM, and CatBoost models were evaluated and compared. To improve the detection of default cases, cost-sensitive learning was incorporated and compared with standard model versions. While accuracy and AUC changed very little, class weighting increased the recall of the default class from below 5% to around 62%. Among all models, the tuned gradient boosting methods achieved the best results (AUC $\approx$ 0.66, F1 $\approx$ 0.41). SHAP analysis showed that the most influential features were related to credit history, loan characteristics, and borrower risk.

Keywords—loan default prediction; imbalanced classification; cost-sensitive learning; gradient boosting; SHAP.

I. INTRODUCTION

Vehicle financing accounts for a substantial share of consumer lending, and anticipating whether a borrower will default is essential for managing credit risk and limiting financial loss. As lending portfolios grow, manual or rule-based assessment becomes impractical, which motivates data-driven models that learn default patterns from historical borrower and loan characteristics.

A central difficulty in default prediction is class imbalance: because defaulters form only a small fraction of borrowers, a model can achieve high accuracy simply by predicting that no one defaults, while failing to identify the at-risk borrowers who matter most. Evaluation must therefore go beyond accuracy and consider sensitivity, specificity, Cohen's

kappa, ROC/AUC, and the F1-score. Prior work has shown that tree-based ensembles, and gradient boosting frameworks in particular XGBoost [1], LightGBM [2], and CatBoost [3] are highly effective on heterogeneous tabular data. Class imbalance is commonly addressed through resampling techniques such as SMOTE [4] or through cost-sensitive learning, while SHAP [5] offers a unified framework for interpreting model predictions.

This study applies several machine learning methods to a large vehicle loan dataset with a highly imbalanced target variable. The analysis includes data preprocessing, feature engineering, model comparison, hyperparameter tuning, model interpretation using SHAP, and borrower segmentation through k-means clustering. In addition, the impact of cost-sensitive learning is examined by comparing weighted and unweighted versions of the models.

II. METHODOLOGY

A. Dataset Description

The dataset consists of 233,154 vehicle loan records described by 41 original variables. After feature engineering, 38 features were retained for modeling. Table I provides a summary of the dataset and the target variable. The response variable, *loan_default*, is binary and highly imbalanced, with only 21.71% of borrowers belonging to the default class (Fig. 1). Because of this imbalance, model performance cannot be evaluated using accuracy alone and requires additional metrics that better reflect minority-class detection. Missing values were present only in the employment-type variable (3.29% of observations) and were replaced with the most frequent category.

TABLE I
OVERVIEW OF DATASET

Item	Value
Raw dataset shape	233,154 × 41
Features after engineering	38 columns
Target variable	loan_default
Negative class: No default (0)	182,543 (78.29%)
Positive class: Default (1)	50,611 (21.71%)
Missing values	Employment.Type only (3.29%)

The variables can be grouped into several categories, including borrower demographics, loan and asset characteristics, identity-verification indicators, credit-bureau information, and identifier variables, as summarized in Table II.

TABLE II
GROUPING OF VARIABLES BY CATEGORY

Variable Group	Variables
Demographics	Date.of.Birth, Employment.Type
Loan / asset	disbursed_amount, asset_cost, ltv
Identity flags	Aadhar, PAN, VoterID, Driving, Passport, MobileNo
Credit bureau – score	PERFORM_CNS.SCORE, PERFORM_CNS.SCORE.DESCRPTION
Credit bureau – primary accts	PRI.NO/ACTIVE/OVERDUE.ACCTS, PRI.CURRENT.BALANCE, PRI.SANCTIONED/DISBURSED.AMOUNT
Credit bureau – secondary accts	SEC.* equivalents, PRIMARY/SEC.INSTAL.AMT
Credit bureau – activity	AVERAGE.ACCT.AGE, CREDIT.HISTORY.LENGTH, NEW/DELINQUENT.ACCTS.IN.LAST.SIX.MONTHS, NO.OF_INQUIRIES
Identifiers	branch_id, supplier_id, manufacturer_id, State_ID, pincode/employee codes

Table III presents descriptive statistics for the main numerical features. The loan amount and asset cost distributions are right-skewed, with mean values exceeding their medians and a small number of very large observations. The median credit score is zero, suggesting that a substantial proportion of borrowers have no recorded bureau history. After correcting the date-parsing issue described in the preprocessing stage, borrower age at disbursal ranged from 18 to 69 years.

TABLE III
DESCRIPTIVE STATISTICS OF KEY NUMERICAL FEATURES

Feature	Mean	Std	Median	Min	Max
Loan Amount	54,357	12,971	53,803	13,320	990,572
Asset Cost	75,865	18,945	70,946	37,000	1,628,992
LTV	74.75	11.46	76.80	10.03	95.00
Credit Score	289.46	338.37	0.00	0.00	890.00
Age	34.47	9.82	32.67	18.00	69.13

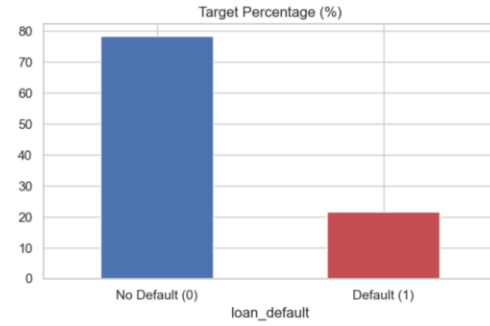


Fig. 1. Class distribution of the loan default target

B. Data Preprocessing and Feature Engineering

Several preprocessing steps were performed before modeling. Missing values appeared only in the employment-type variable and were imputed with the most frequent category. Borrower age at disbursal was calculated from the date of birth and the disbursal date. During data exploration, a date-parsing issue was detected: birth years between 1950 and 1968 had been incorrectly mapped to the 2050s and 2060s, producing impossible negative ages. After applying a century correction, the age range became 18 to 69 years. Credit-history variables stored as text (e.g., "1yrs 11mon") were converted into total months.

Several additional features were created to capture credit behavior. These included a bureau-history indicator, active-account and overdue-account ratios, and a risk-group variable derived from the credit-score description. To handle the high-cardinality branch identifier, frequency encoding was used instead of one-hot encoding. Finally, redundant variables, transformed date/text fields, and high-cardinality identifiers such as supplier, pincode, and employee codes were removed.

C. Exploratory Data Analysis

Exploratory analysis was conducted to examine the characteristics of the data and their relationship with the target variable. Among the numerical predictors,

defaulting borrowers tended to have slightly higher loan-to-value ratios and were somewhat younger, although no single variable clearly separated the two classes. The credit score showed a bimodal distribution (Fig. 3): many borrowers had a score of zero, indicating no bureau history, while the remaining observations were concentrated at higher scores.

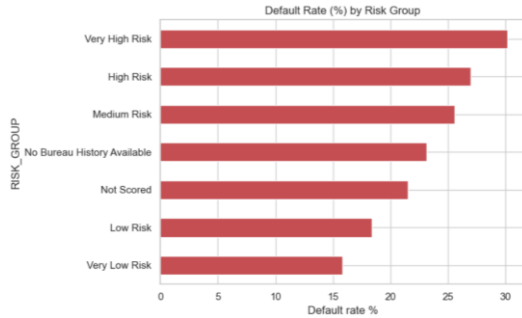


Fig. 2. Default rate by risk group.

The categorical variables showed stronger patterns. The default rate increased steadily across risk groups, from about 16% in the lowest-risk category to about 30% in the highest-risk category (Fig. 2). Self-employed borrowers also exhibited a slightly higher default rate than salaried borrowers (23% versus 20%). Correlation analysis indicated only weak-to-moderate relationships among the numerical features and no evidence of severe multicollinearity.

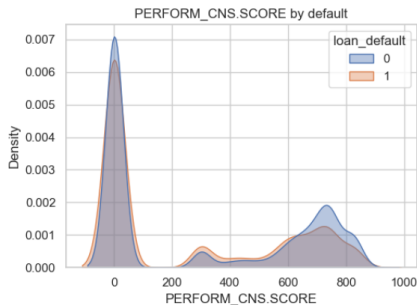


Fig. 3. Distribution of the credit score (PERFORM_CNS.SCORE) by default class.

D. Encoding & Splitting

Before modeling, the features were prepared for the machine learning algorithms. Numerical variables were standardized, while categorical variables such as employment type, risk group, state, and manufacturer were one-hot encoded. The branch identifier had already been transformed using frequency encoding because of its high cardinality. To avoid data leakage, all preprocessing steps were fitted only on the training data within the cross-validation pipeline. Finally, the data were split into 80% training and 20% test sets

using stratified sampling with a fixed random seed, preserving the same default rate in both sets.

E. Feature Selection

Feature selection was performed using two different approaches: mutual information and random-forest feature importance. The rankings obtained from these methods were averaged to create a combined ranking. As shown in Table IV, both methods identified credit score, loan amount, age at disbursal, branch frequency, and loan-to-value ratio as some of the most informative features. In contrast, most sparse one-hot encoded identifier variables appeared near the bottom of the ranking.

No single feature appears to dominate prediction. Both methods highlighted similar variables, suggesting that these features contain useful information for predicting default.

TABLE IV

TOP-RANKED FEATURES BY MUTUAL INFORMATION AND RANDOM FOREST IMPORTANCE

Feature	Mutual Info	RF Importance	Avg. Rank
Credit Score	0.013	0.057	3.0
Loan Amount	0.007	0.089	4.0
Age	0.006	0.062	5.5
Branch Frequency	0.008	0.044	5.5
Primary Disbursed Amt.	0.007	0.043	6.5
Overdue Ratio	0.009	0.023	8.5
LTV	0.004	0.098	9.5
Primary Sanctioned Amt.	0.005	0.045	10.5

F. Predictive Modeling

Several supervised learning models were evaluated and compared. These included logistic regression, decision tree, support vector machine, random forest, XGBoost, LightGBM, and CatBoost. These span linear, tree-based, and ensemble methods.

Model performance was first assessed using five-fold stratified cross-validation on the training data, which preserved the class distribution in each fold. Each model was evaluated in both unweighted and cost-sensitive settings. The best-performing models were then further improved through hyperparameter tuning.

Because the support vector machine performed poorly in the baseline comparison, it was not included in the hyperparameter tuning stage, leaving six models for the tuned comparison. A separate test set, not used during training or model selection, was retained for the final evaluation presented in the Results section.

G. Class Imbalance Handling

Since only 21.71% of the observations correspond to defaults, models trained without any adjustment tend to favor the majority class and identify very few defaulters. To address this issue, cost-sensitive learning was used instead of resampling. Class weights inversely proportional to class frequencies were applied to logistic regression, decision tree, support vector machine, and random forest models. For XGBoost and LightGBM, the scale-positive-weight parameter was set to the ratio of negative to positive cases (approximately 3.6), while CatBoost used its built-in balanced weighting option.

Each model was trained in both unweighted and cost-sensitive versions, and their performances were compared using the same evaluation metrics.

H. Hyperparameter Optimization

The six main models were tuned using grid search with three-fold cross-validation on the training data. The F1-score was used as the optimization metric because it considers both precision and recall, which are particularly important for the minority class in an imbalanced dataset. Table V summarizes the parameter values tested for each model and the configuration that achieved the best cross-validated F1-score.

TABLE V
SEARCHED HYPERPARAMETER GRIDS AND BEST CONFIGURATIONS

Model	Parameters Searched	Best
Logistic Reg.	C: {0.01, 0.1, 1, 10}	C = 1
Decision Tree	max_depth: {4, 6, 8, 10}; min_leaf: {50, 100, 200}	depth = 8; min_leaf = 200
Random Forest	n_est: {150, 250}; max_depth: {8, 12}; min_leaf: {10, 50}	depth = 12; min_leaf = 50; n_est = 150
XGBoost	n_est: {200, 400}; lr: {0.03, 0.05, 0.1}; max_depth: {3, 5}	lr = 0.1; depth = 3; n_est = 400
LightGBM	n_est: {200, 400}; lr: {0.03, 0.05, 0.1}; leaves: {31, 63}	lr = 0.05; n_est = 200; leaves = 31
CatBoost	iter: {200, 400}; lr: {0.03, 0.05, 0.1}; depth: {4, 6}	depth = 6; iter = 400; lr = 0.1

I. Model Evaluation and Interpretation

Model performance was evaluated using accuracy, precision, recall, specificity, F1-score, ROC-AUC, and Cohen's kappa. Particular attention was given to recall and F1-score because identifying default cases was more important than maximizing overall accuracy. ROC and precision-recall curves were also used to examine the trade-off between detecting defaulters and generating false positives. To compare model predictions statistically, McNemar's test was applied to the tuned models on the test set.

To better understand the final model, SHAP (SHapley Additive exPlanations) was applied to the best-performing tuned model. SHAP was used to measure the contribution of each feature to the predictions and to identify the variables with the greatest overall impact. This helped determine whether the model's decisions were consistent with established knowledge of credit risk.

III. RESULTS

A. Model Comparison and the Effect of Imbalance Handling

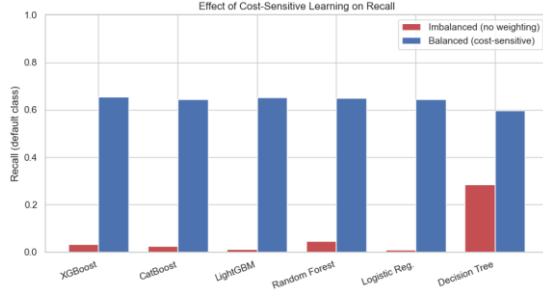


Fig. 4. Recall on the default class before and after cost-sensitive learning, for each tuned model.

The class imbalance had a major impact on model performance. Without any adjustment, most models achieved an accuracy of around 78%, but their recall for the default class remained below 5%, meaning that very few defaulters were correctly identified. Cohen's kappa values were also close to zero. After applying cost-sensitive learning, recall increased substantially, reaching about 62–65% across the models (Fig. 4). This improvement came with lower precision and overall accuracy, which was expected. ROC-AUC values changed very little, suggesting that class weighting mainly affected the classification threshold rather than the model's ability to rank observations. These results highlight why accuracy alone is not an adequate measure for imbalanced classification problems.

TABLE VI

PERFORMANCE OF TUNED MODELS ON THE TEST SET

Model	Acc.	Prec.	Recall	F1	AUC	Spec.	Kappa
XGBoost	0.598	0.303	0.655	0.414	0.663	0.582	0.167
CatBoost	0.602	0.303	0.644	0.413	0.663	0.590	0.167
LightGBM	0.594	0.300	0.651	0.411	0.661	0.579	0.162
Random Forest	0.583	0.293	0.649	0.404	0.649	0.565	0.149
Logistic Reg.	0.583	0.292	0.644	0.401	0.644	0.566	0.146
Decision Tree	0.590	0.286	0.596	0.387	0.624	0.589	0.133

Table VI summarizes the performance of the tuned models on the test set. The three gradient boosting

methods achieved the strongest results and performed very similarly. XGBoost and CatBoost obtained the highest F1-scores (0.414 and 0.413) and kappa values (0.167), followed closely by LightGBM. The remaining models showed slightly lower performance, while the decision tree produced the weakest results.

The effect of hyperparameter tuning varied across models. For the boosting methods, tuning led to only small improvements because their initial performance was already strong. In contrast, tuning had a substantial impact on the random forest model. The untuned balanced random forest achieved an F1-score below 0.08, but limiting tree depth during tuning increased its F1-score to 0.40, bringing it close to the performance of the boosting models. This suggests that class weighting alone may not always be sufficient and can benefit from additional model regularization.

B. Statistical Significance

McNemar's test was used to compare the test-set predictions of the best-performing model (XGBoost) with those of the other tuned models. The results are presented in Table VII. All pairwise comparisons were statistically significant at the 0.05 level. Given the large test set ($\approx 46,000$ observations), even small performance differences can become statistically significant. Therefore, practical differences are more informative than p-values alone.

XGBoost performed better than logistic regression, decision tree, random forest, and LightGBM in terms of correctly classified observations. Although the comparison between XGBoost and CatBoost was also statistically significant, the practical difference was very small. The two models disagreed on only about 6% of the test cases and showed nearly identical overall performance. Therefore, XGBoost and CatBoost can be considered effectively equivalent in practice. This result illustrates the difference between statistical significance and practical significance when evaluating models on large datasets.

TABLE VII

MCNEMAR TEST RESULTS: XGBOOST VS. OTHER TUNED MODELS

Comparison	b	c	χ^2	p	Sig.
Logistic Reg.	3612	2917	73.77	<0.001	Yes
Decision Tree	4385	4019	15.85	<0.001	Yes
Random Forest	3169	2488	81.74	<0.001	Yes
LightGBM	1451	1286	9.83	0.002	Yes
CatBoost	1371	1540	9.70	0.002	Yes

C. ROC and Precision–Recall Analysis

The performance of the best model is illustrated in Figs. 5 and 6. The ROC curve (Fig. 5) lies above the diagonal reference line and achieves an AUC of 0.663, indicating a moderate ability to distinguish between defaulters and non-defaulters. The precision-recall curve (Fig. 6), which is particularly useful for imbalanced datasets, remains above the no-skill baseline of 0.22 and achieves an average precision of 0.336. As expected, precision decreases as recall increases, reflecting the trade-off between identifying more defaulters and generating additional false positives. Overall, the model separates defaulters from non-defaulters better than chance, although precision remains limited because defaults are relatively rare.

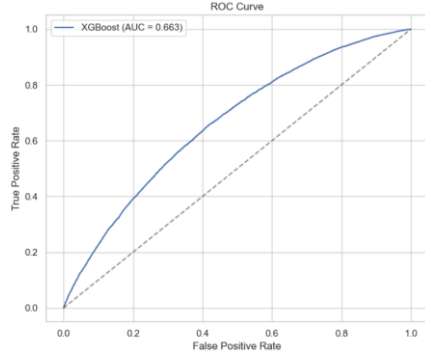


Fig. 5. ROC curve for the best tuned model (XGBoost).

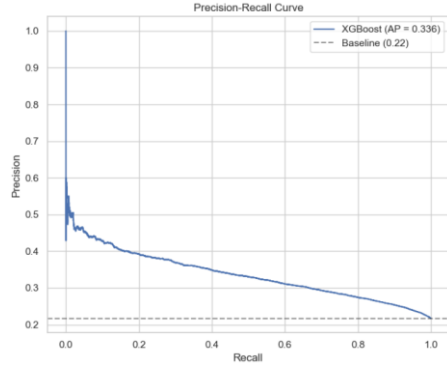


Fig. 6. Precision-recall curve for the best tuned model (XGBoost).

D. Feature Importance and Model Interpretation (SHAP)

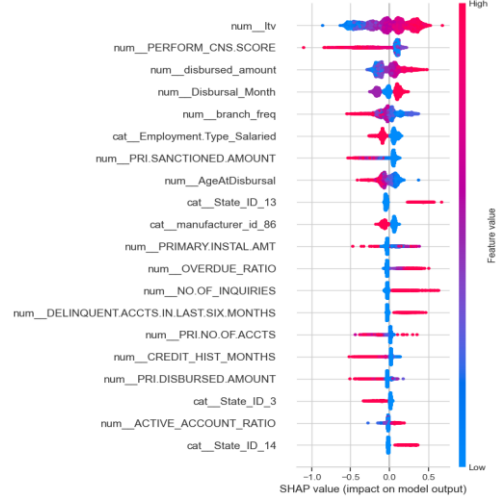


Fig. 7. SHAP summary plot showing the global impact of each feature on the model's output.

Fig. 7 shows the SHAP summary for the best-performing model. The most influential features were the loan-to-value ratio, credit score, loan amount, disbursement month, and branch frequency, followed by employment type, age, and several credit-behavior variables. Higher loan-to-value ratios and younger ages were generally associated with a higher probability of default, while higher credit scores and salaried employment were associated with lower risk. Similarly, borrowers with more recent credit inquiries and overdue accounts tended to have higher predicted default probabilities.

Many of the features highlighted by SHAP were also ranked highly in the feature-selection analysis, including credit score, loan amount, age, branch frequency, and loan-to-value ratio. This consistency provides additional confidence in the model's interpretation.

E. Unsupervised Borrower Segmentation

TABLE VIII

PROFILES OF THE TWO BORROWER CLUSTERS

Feature	Cluster 0	Cluster 1
Size	119,185	113,969
Credit Score	0.8	591.3
Active Account Ratio	0.00	0.55
Overdue Ratio	0.00	0.09
Age	32.6	36.4
LTV	73.6	76.0
Default Rate (%)	23.1	20.3

To further explore the borrower population, k-means clustering was applied to the encoded features. The elbow method did not indicate a clear optimal number of clusters, so two clusters were selected for easier interpretation. As shown in Table VIII, the clusters differ mainly in their credit-history characteristics, while their loan terms are quite similar. Cluster 0 consists of borrowers with little or no credit history: their credit score is close to zero, they have almost no active or overdue accounts, and they are slightly younger. Cluster 1 contains borrowers with an established credit record, including higher credit scores, active accounts, and some overdue-account history. Since loan amount and loan-to-value ratio are similar across both groups, the clustering appears to be driven primarily by credit-history information.

The thin-file cluster has a higher default rate (23.1%) than the established-history cluster (20.3%). This suggests that borrowers without a documented credit history may represent a somewhat riskier group. The finding is also consistent with the bimodal credit-score distribution observed during exploratory analysis. The clustering results reinforce the importance of credit-history characteristics in explaining default risk.

IV. CONCLUSION

This study applied a complete machine learning pipeline to vehicle loan default prediction on a large, highly imbalanced dataset. Among the tuned models, XGBoost and CatBoost performed best and were effectively equivalent ($F1 \approx 0.41$, $AUC \approx 0.66$, $\kappa \approx 0.17$). Several findings stand out. Accuracy proved misleading under imbalance: unadjusted models reached about 78% accuracy yet detected almost no

defaulters, whereas cost-sensitive learning raised recall from below 5% to roughly 62% with little change in AUC. Class weighting alone was not always sufficient — the random forest improved only after its tree depth was constrained, indicating that imbalance handling may need to be combined with regularization. SHAP identified the loan-to-value ratio, credit score, age, and credit-behavior variables as the strongest predictors, consistent with the feature-selection results, and k-means clustering separated borrowers mainly by credit history, with thin-file customers defaulting more often.

The main limitation is the moderate AUC, which reflects the intrinsic difficulty of the problem, since no single feature is strongly predictive. In addition, branch frequency encoding was fitted on the full dataset; though target-independent, a fold-wise encoding would be more rigorous. Future work could explore resampling methods such as SMOTE, per-model threshold tuning, additional features, and validation on external datasets.

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